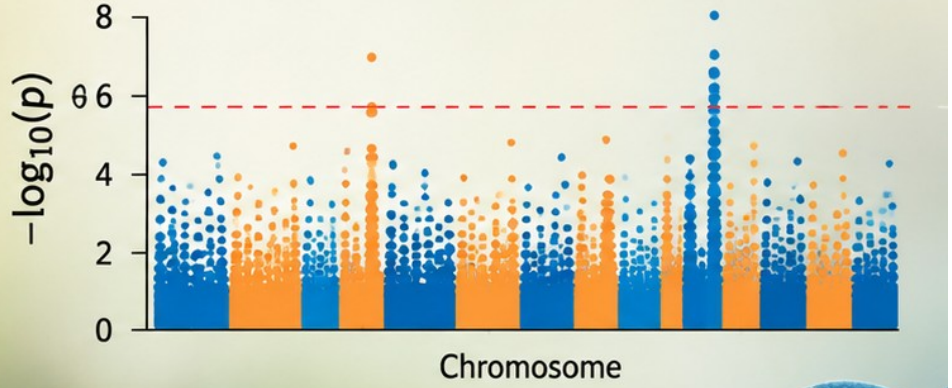
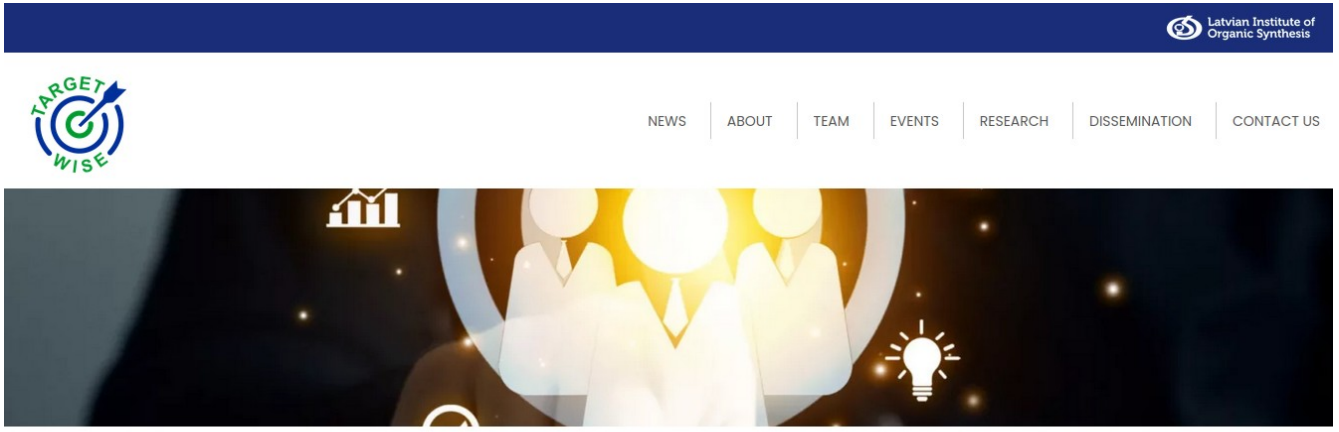
Genome-Wide Association Studies (GWAS) for Evolutionary Biology

Nikolay Oskolkov, Group Leader (PI) at NIRI, Riga, Latvia





Metabolic Research Group at NIRI



1 more postdoctoral fellow and 1 PhD student to be hired

If you know anyone who might be interested, please contact me!



PhD. Nikolay Oskolkov
Group Leader (PI) of the Metabolic Research Group



Mohammad Banabazi, postdoc in Metabolic Research Group



Kristina Grausa, PhD student in Metabolic Research Group



Daniel Rivas, MD, postdoc in Metabolic Research Group

About us

Organizer: Oliver Hooker, PR Stats courses

oliver@prstats.org

Instructor:

Dr. Nikolay Oskolkov, MRG Group Leader, NIRI, Riga,
lecturer at Lund University, Lund,Sweden

nikolay.oskolkov@niri.lv



@NikolayOskolkov

@oskolkov.bsky.social

Personal homepage:
https://nikolay-oskolkov.com

Brief introduction: who am I

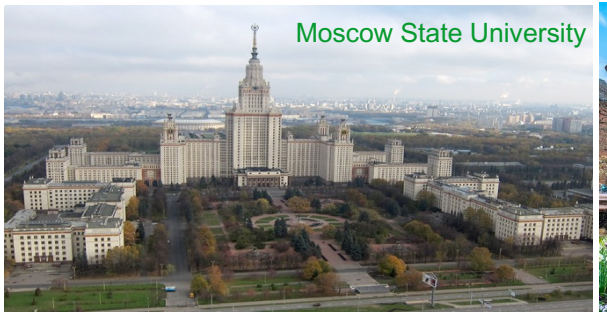
2007 PhD in theoretical physics in Moscow, Russia

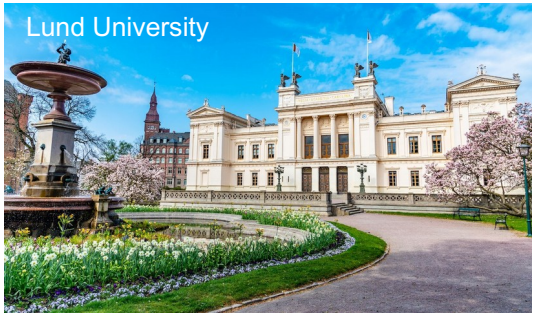
2011 medical genomics at Lund University, Sweden

2016 bioinformatician at NBIS SciLifeLab, Sweden

2025 Metabolic Research Group leader, NIRI, Latvia

60+ publications; h-index = 28; 4,600+ citations

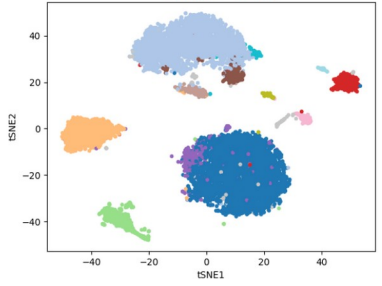








Single cell



Biomedical data integration

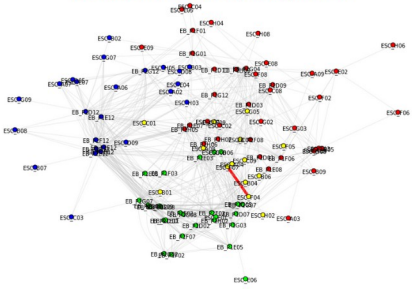
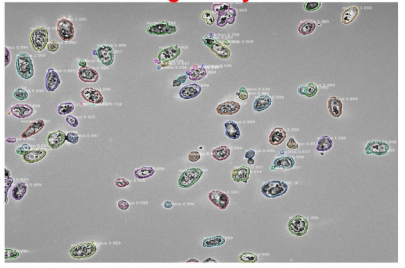
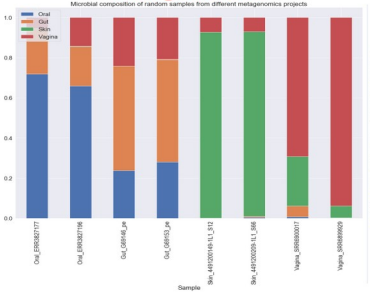


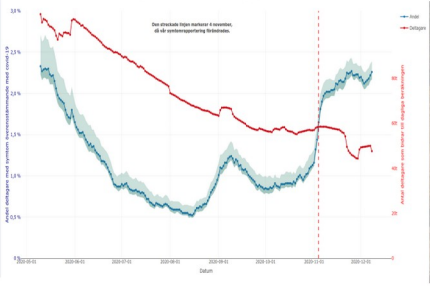
Image analysis



Metagenomics



Epidemiology



Ancient microbiome

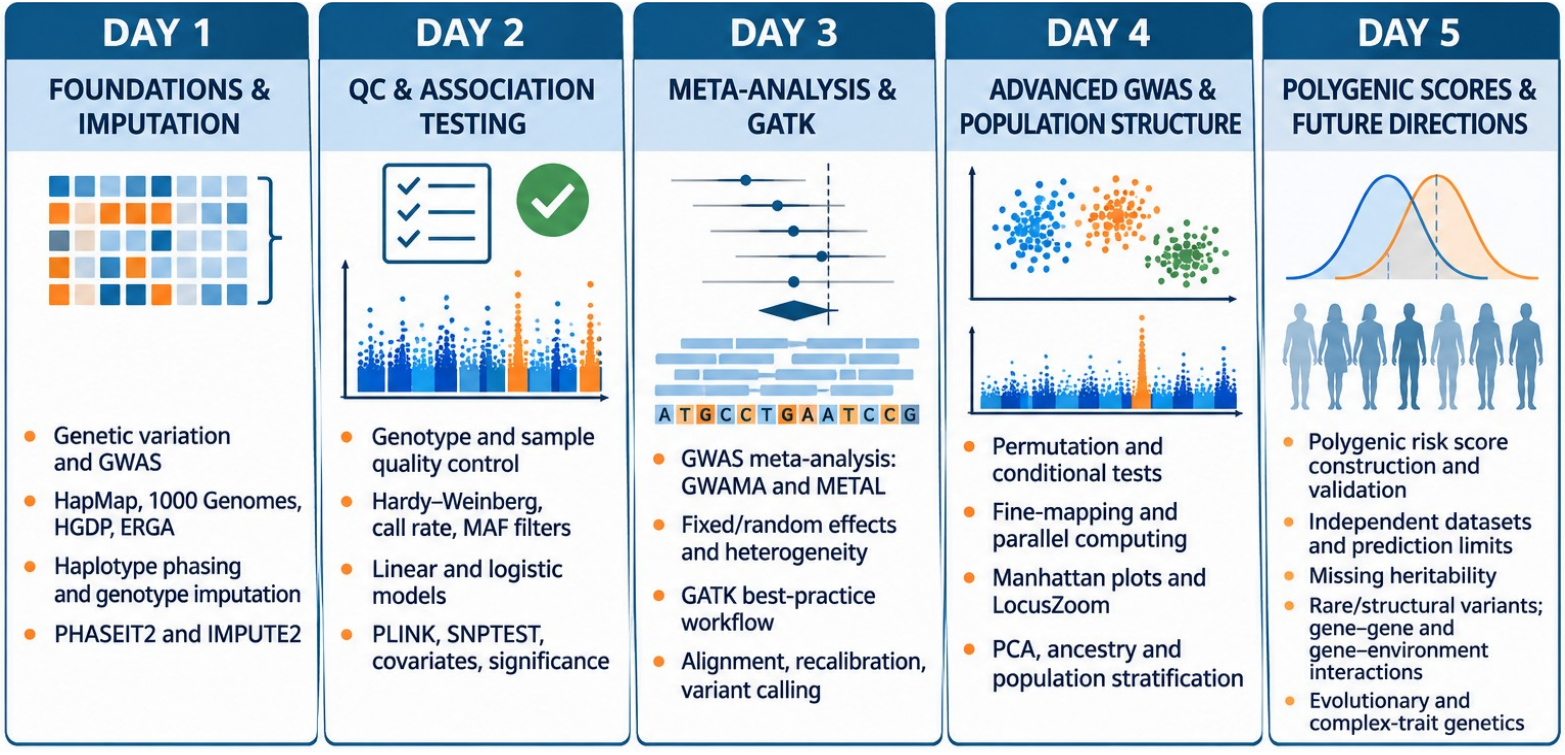


About you

- Name
- University / Institute / Company
- Research interest(s)
- Previous experience with computational analysis and bioinformatics
- Motivation to join the course
- Expectations from the course

Course outline

Day 1: introduction to genomics data analysis, genome-wide association, imputation
Day 2: quality control of genomics data and genome-wide association statistical test
Day 3: meta-analysis of GWAS studies, GATK workflow, alignment, variant calling
Day 4: PCA and adjusting for population structure, permutations, conditional analysis
Day 5: polygenic risk score prediction, missing heritability, Mendelian Randomization



Course program

Monday 9th – Classes from 15:00-21:00

Session 1 – 02:30:00 – Introduction to Genetic Variation and GWAS
Overview of genome-wide association studies; genetic variation; major genomic reference resources including the International HapMap Project, 1000 Genomes Project, Human Genome Diversity Project, and European Reference Genome Atlas.

Break – 01:00:00

Session 2 – 02:30:00 – Genotype Phasing and Imputation
Principles of haplotype phasing and genotype imputation; practical workflows using PHASEIT2 and IMPUTE2; improving genomic coverage and statistical power.

Tuesday 10th – Classes from 15:00-21:00

Session 3 – 02:30:00 – Quality Control of Genotype Data
Filtering genotyping array data; Hardy-Weinberg equilibrium tests, SNP call-rate thresholds, minor allele frequency filters, and identification of problematic samples.

Break – 01:00:00

Session 4 – 02:30:00 – Association Testing with PLINK and SNPTEST
Linear and logistic association models; covariate adjustment; interpretation of association statistics; genome-wide significance thresholds.

Wednesday 11th – Classes from 15:00-21:00

Session 5 – 02:30:00 – Meta-Analysis of GWAS Results
Combining association results across studies using GWAMA and METAL; fixed-effect and random-effect models; heterogeneity assessment.

Break – 01:00:00

Session 6 – 02:30:00 – Genomic Data Processing with GATK
GATK best-practice workflow including quality control, sequence alignment, duplicate handling, base quality score recalibration, indel realignment, and variant calling with HaplotypeCaller.

Thursday 12th – Classes from 15:00-21:00

Session 7 – 02:30:00 – Advanced Association Testing and Computational Strategies
Permutation testing, conditional association analyses, fine-mapping concepts, and strategies for parallelizing large-scale GWAS computations.

Break – 01:00:00

Session 8 – 02:30:00 – Visualization and Population Structure Analysis
Visualization of GWAS results using Manhattan plots and LocusZoom; population structure correction using PCA; identification of confounding due to ancestry.

Friday 13th - Classes from 15:00-21:00

Session 9 – 02:30:00 – Polygenic Risk Scores
Construction, optimization, and validation of polygenic risk scores; evaluation on independent datasets; applications and limitations.

Break – 01:00:00

Session 10 – 02:30:00 – Missing Heritability and Future Directions in GWAS
The missing heritability problem; rare variants, structural variation, gene-gene interactions, gene-environment interactions, multi-omics integration, and future directions in complex trait genetics and evolutionary biology.

Housekeeping rules

- Please ask your questions anytime, it is ok to interrupt me, or you can raise your hand and ask. You can also ask questions online in the zoom chat or later by email.

- I kindly ask you to please keep your webcams on for better communication, this is of course not mandatory an up to you but will help me to adjust me explanations

- The course includes 10 lectures (~1 h each) and 10 practicals (~1.5 h each), there is a 15 min break after each lecture and 1 h lunch break every day of the course. We can potentially make longer breaks if needed.

- During practicals, we will use Rmarkdown and Jupyter notebooks. At the end of each practical, I will use rendered html-versions of the notebooks to go through command lines with my explanations.

- The material is based on data and problems from computational biology, however the concepts discussed are general and can be applied for other types of data.

Practical information: time and github repo

The course will take place from 2 pm to 8 pm UK time (3 pm to 9 pm CET)

The course GitHub repository containing lectures and exercises is:

https://github.com/NikolayOskolkov/GWAS

Please bookmark this address!

GWAS Public

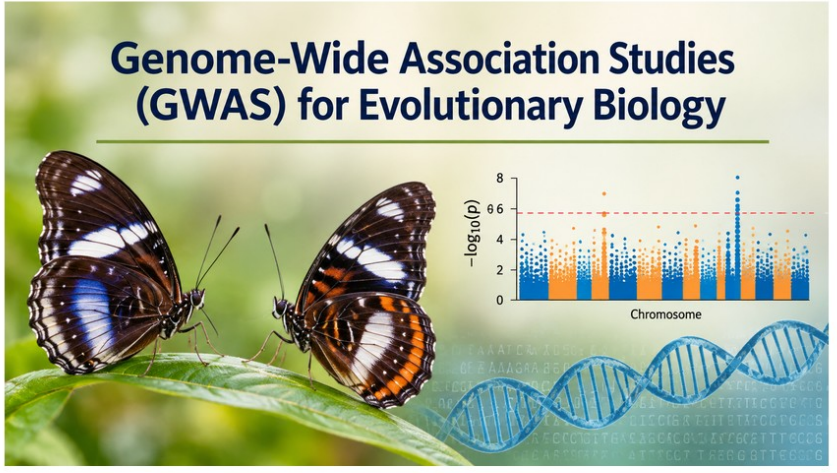
Pin Watch 0 Fork 0 Star 0

main 1 Branch 0 Tags Go to file T Add file Code

NikolayOskolkov edited README 2a55d6f · 4 minutes ago 4 Commits

images	added course logo	yesterday
README.md	edited README	4 minutes ago

README



Genome-Wide Association Studies (GWAS) for Evolutionary Biology

Instructor

- Dr. Nikolay Oskolkov, Group Leader (PI) at NIRI, Riga, Latvia

About

This repository contains material for GWAS course

Readme
Activity
0 stars
0 watching
0 forks

Releases

No releases published
Create a new release

Packages

No packages published
Publish your first package

Contributors 1

NikolayOskolkov Nikolay Oskolkov

Google Colab

Python based



Python
(with rich ecosystem)

Google Drive
(easy file access)

Free GPU/TPU
(for ML and deep learning)

No installation
(runs in the browser)

VS

Post Cloud

RStudio and R based



R
(statistical computing)

RStudio
(full IDE)

Dedicated servers
(stable and secure)

Large storage
(for big datasets)

Key strengths

- Easy to get started (no setup)
- Great for Python and machine learning
- Free access to GPUs/TPUs
- Seamless integration with Google Drive
- Ideal for notebooks and reproducible workflows

Limitations

- Python only (no RStudio)
- Session time limits (can disconnect)
- Less control over the computing environment
- Not ideal for large-scale or long-running jobs

Key strengths

- RStudio with full IDE and packages
- Great for statistical analysis and visualization
- Stable, dedicated computing environment
- Suitable for large datasets and long-running analyses
- Supports collaborative and reproducible workflows

Limitations

- R only (no Python)
- No free GPU/TPU access (CPU only)
- Requires access to Post Cloud infrastructure
- Slightly more setup and learning curve for beginners

Both platforms support your research – choose the one that fits your needs

Google Colab for flexible, Python-based, cloud computing | Post Cloud for RStudio, R and dedicated resources

posit.cloud/content/13008494

Account owners can now enable **Posit Assistant**, an AI coding assistant for RStudio projects. Get started quickly with a free trial Posit AI subscription. **Learn more.**

Posit Cloud

Your Workspace / PopGen_PopStructure

Nikolay Oskolkov

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File Edit Code View Plots Session Build Debug Profile Tools Help

Untitled1*

```
gen <- t(matrix(c(1, 0, 2, 0, 2, 0, 2, 1, 1, 1, 0, 1, 0, 2, 1, 2, 1, 1, 1, 1, 1, 0, 1, 0, 2, 0, 1, 1, 0, 2
colnames(gen) <- paste0("Ind", 1:5)
rownames(gen) <- paste0("SNP", 1:7)
print(gen)

gen_centered <- scale(gen, center = TRUE, scale = FALSE) # mean centering
covariance <- t(gen_centered) %*% gen_centered # computing variance-covariance matrix
eig <- eigen(covariance) # performing eigen value decomposition of the variance-covariance matrix

barplot(eig$values / sum(eig$values), names = paste0("PC", 1:5))

```

11:1 (Top Level) R Script

Environment History Connections Tutorial

Import Dataset 143 MiB List

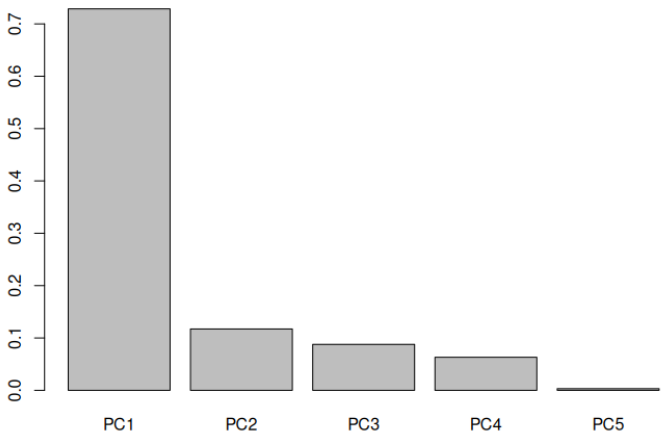
R Global Environment

Data

covariance	num [1:5, 1:5] 6 3 -1 -3 -4 ...
eig	List of 2
gen	num [1:7, 1:5] 1 0 2 0 2 0 2 1 1 1 ...
gen_centered	num [1:7, 1:5] 0 -1 1 -1 1 ...

Files Plots Packages Help Viewer Presentation

Zoom Export Publish



Console Terminal Background Jobs

R 4.6.1 · /cloud/project/

```
Type 'license()' or 'licence()' for distribution details.

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

Session restored from your saved work on 2026-Sep-27 14:27:56 UTC (3 hours ago)
> gen <- t(matrix(c(1, 0, 2, 0, 2, 0, 2, 1, 1, 1, 0, 1, 0, 2, 1, 2, 1, 1, 1, 1, 1, 0, 1, 0, 2, 0, 1, 1, 0, 2,
1, 2, 0, 1, 0), 5, by = TRUE))
> colnames(gen) <- paste0("Ind", 1:5)
> rownames(gen) <- paste0("SNP", 1:7)
> print(gen)
     Ind1 Ind2 Ind3 Ind4 Ind5
SNP1    1    1    1    0    0
SNP2    0    1    2    1    2
SNP3    2    1    1    0    1
SNP4    0    0    1    2    2
SNP5    2    1    1    0    0
SNP6    0    0    1    1    1
SNP7    2    2    1    1    0
> gen_centered <- scale(gen, center = TRUE, scale = FALSE) # mean centering
> covariance <- t(gen_centered) %*% gen_centered # computing variance-covariance matrix
> eig <- eigen(covariance) # performing eigen value decomposition of the variance-covariance matrix
> barplot(eig$values / sum(eig$values), names = paste0("PC", 1:5))
>
```

To run Jupyter notebooks on Google Colab please do the following:

1. Clone this repo: https://github.com/NikolayOskolkov/GWAS
2. Upload the repo to your Google drive
3. Start the Jupyter notebook from the Google drive and mount your google drive:

```
#Mount Google Drive to Colab

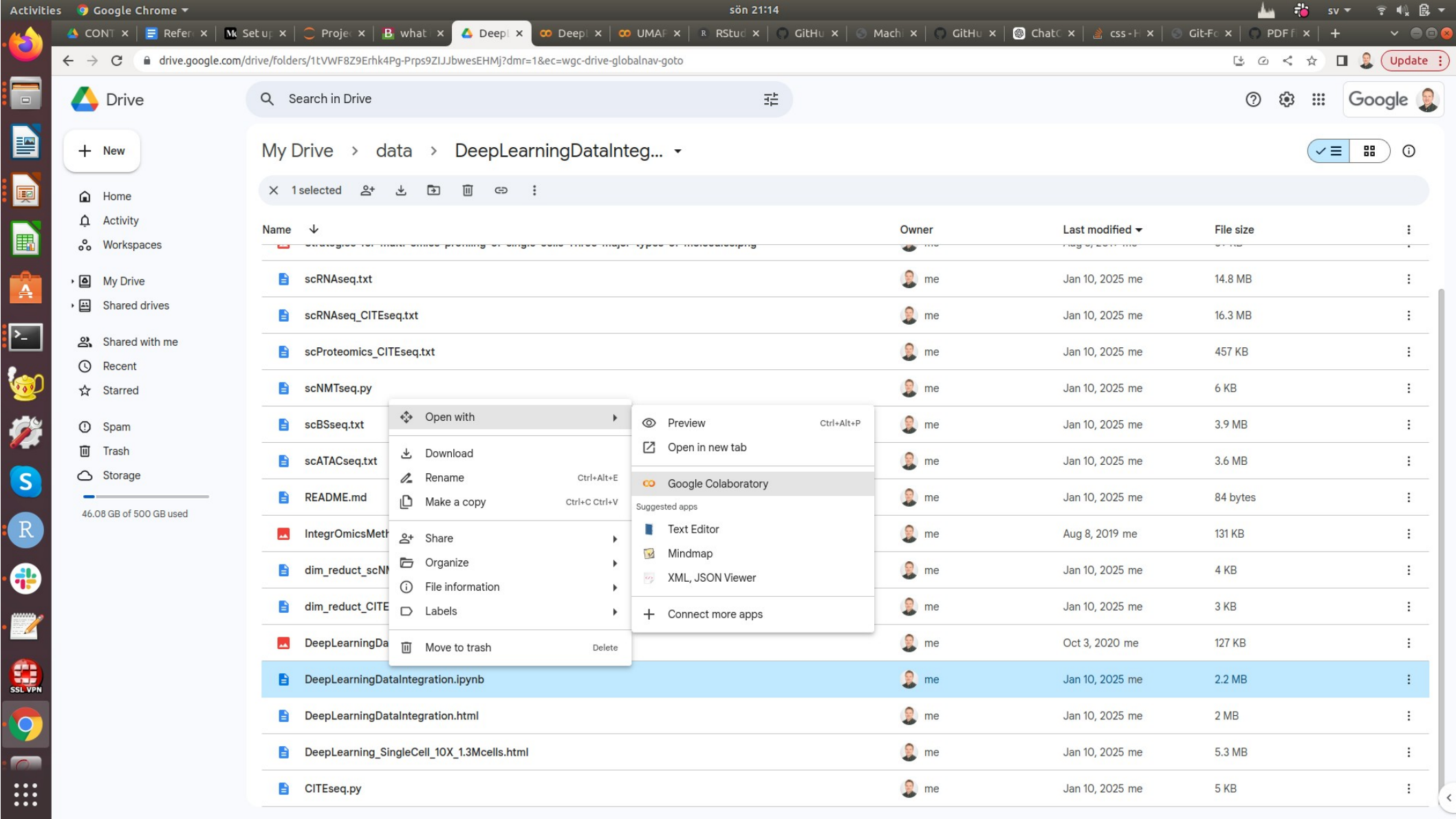
from google.colab import drive
drive.mount('/content/drive')

!ls -l /content/drive/MyDrive/

import os
os.chdir("/content/drive/MyDrive/data")
```

N.B. you need a Google account to be able to run Google Colab!




Activities
Google Chrome
sön 21:14
drive.google.com/drive/folders/1tVWF8Z9Erhk4Pg-Prps9ZIJJbwesEHMj?dmr=1&ec=wgc-drive-globalnav-goto
Update
Drive
Search in Drive
Google
New
Home
Activity
Workspaces
My Drive
Shared drives
Shared with me
Recent
Starred
Spam
Trash
Storage
46.08 GB of 500 GB used
My Drive › data › DeepLearningDataInteg...
1 selected
Name
Owner
Last modified
File size
scRNAseq.txt me Jan 10, 2025 me 14.8 MB
scRNAseq_CITEseq.txt me Jan 10, 2025 me 16.3 MB
scProteomics_CITEseq.txt me Jan 10, 2025 me 457 KB
scNMTseq.py me Jan 10, 2025 me 6 KB
scBSseq.txt me Jan 10, 2025 me 3.9 MB
scATACseq.txt me Jan 10, 2025 me 3.6 MB
README.md me Jan 10, 2025 me 84 bytes
IntegrOmicsMeth me Aug 8, 2019 me 131 KB
dim_reduct_scN me Jan 10, 2025 me 4 KB
dim_reduct_CITE me Jan 10, 2025 me 3 KB
DeepLearningDa me Oct 3, 2020 me 127 KB
DeepLearningDataIntegration.ipynb me Jan 10, 2025 me 2.2 MB
DeepLearningDataIntegration.html me Jan 10, 2025 me 2 MB
DeepLearning_SingleCell_10X_1.3Mcells.html me Jan 10, 2025 me 5.3 MB
CITEseq.py me Jan 10, 2025 me 5 KB
Open with
Download
Rename Ctrl+Alt+E
Make a copy Ctrl+C Ctrl+V
Share
Organize
File information
Labels
Move to trash Delete
Preview Ctrl+Alt+P
Open in new tab
Google Colaboratory
Suggested apps
Text Editor
Mindmap
XML, JSON Viewer
Connect more apps

colab.research.google.com/drive/1d9zF92sgQe9e32EXN4qO0fxOGEaiYFie#scrollTo=mBtONdeBZbj6

Colab is now part of Google AI Plans. Get upgraded compute power and a suite of new AI tools all in one bundle. Explore the Plans

ANGSD_Lab_GoogleColab.ipynb

Probabilistic Variant Calling with ANGSD on Low-Coverage Data

Course: Population Genomics and Population Structure Analysis
Instructor: Nikolay Oskolkov
Platform: Google Colab

In this tutorial, we will:

1. Install **Conda**, **ANGSD**, and **pcangsd** in Google Colab.
2. Mount **Google Drive** to access the input BAM files.
3. Prepare a list of BAM files with corrected paths.
4. Run **ANGSD** to compute genotype likelihoods (the `.beagle.gz` file).
5. Run **pcangsd** to perform PCA on the genotype likelihoods.
6. Plot the PCA result in **R**.
7. Run **NGSadmix** to estimate admixture proportions.
8. Plot the admixture bar plot in **R**.

Input data: 435 low-coverage BAM files from the 1000 Genomes Project, located in your Google Drive at `PopGen_PopStructure/1000G/smallbams/`.

1. Install Conda in Google Colab

Google Colab does not ship with Conda by default. We use the `condacolab` package, which installs a lightweight Miniconda environment directly inside the notebook.

> ⚠️ **Important:** After running the cell below, the Colab kernel will **restart automatically**. This is expected — just continue with the next cell once the restart finishes.

[ ]

```
# Install Conda inside Google Colab
!pip install -q condacolab
import condacolab
condacolab.install()
```

```
📢 Announcement 📢
```

Terminal

```
/content# ll
total 16
drwxr-xr-x 1 root root 4096 Sep 16 13:26 ./
drwxr-xr-x 1 root root 4096 Sep 27 17:29 ../
drwxr-xr-x 4 root root 4096 Sep 16 13:26 .config/
drwxr-xr-x 1 root root 4096 Sep 16 13:26 sample_data/
/content# cd sample_data/
/content/sample_data# ll
total 55512
drwxr-xr-x 1 root root     4096 Sep 16 13:26 ./
drwxr-xr-x 1 root root     4096 Sep 16 13:26 ../
-rwxr-xr-x 1 root root     1697 Jan  1  2000 anscombe.json*
-rw-r--r-- 1 root root   301141 Sep 16 13:26 california_housing_test.csv
-rw-r--r-- 1 root root  1706430 Sep 16 13:26 california_housing_train.csv
-rw-r--r-- 1 root root 18289443 Sep 16 13:26 mnist_test.csv
-rw-r--r-- 1 root root 36523880 Sep 16 13:26 mnist_train_small.csv
-rwxr-xr-x 1 root root      962 Jan  1  2000 README.md*
/content/sample_data# head README.md
This directory includes a few sample datasets to get you started.

*   `california_housing_data*.csv` is California housing data from the 1990
 US
    Census; more information is available at:
    https://docs.google.com/document/d/e/2PACX-1vRhYtsvc5eOR2FWNCwaBiKL6suI
OrxJig8LcSBbmCbyYsayia_DvPOOBlXZ4CAlQ5nlDD8kTaIDRwrN/pub

*   `mnist_*.csv` is a small sample of the
    [MNIST database](https://en.wikipedia.org/wiki/MNIST_database), which i
s
    described at: http://yann.lecun.com/exdb/mnist/

/content/sample_data# ll
total 55512
drwxr-xr-x 1 root root     4096 Sep 16 13:26 ./
drwxr-xr-x 1 root root     4096 Sep 16 13:26 ../
-rwxr-xr-x 1 root root     1697 Jan  1  2000 anscombe.json*
-rw-r--r-- 1 root root   301141 Sep 16 13:26 california_housing_test.csv
-rw-r--r-- 1 root root  1706430 Sep 16 13:26 california_housing_train.csv
-rw-r--r-- 1 root root 18289443 Sep 16 13:26 mnist_test.csv
-rw-r--r-- 1 root root 36523880 Sep 16 13:26 mnist_train_small.csv
-rwxr-xr-x 1 root root      962 Jan  1  2000 README.md*
/content/sample_data#
```

```
[0] 0:bash*                                    "d020aca8653c" 17:34 27-Sep-26
```